Mohammadamin Rami

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EDUCATION

École Polytechnique, Télécom Paris, Institut Polytechnique de Paris — Paris, France
Master of Science in Machine Learning, Communications, and Security — Sept 2025 - Sept 2026
GPA: 17.38/20, Distinction: *Mention Très Bien (With Highest Honors)*

Sharif University of Technology — Tehran, Iran
Bachelor of Science in Electrical Engineering — Sept 2019 - July 2024
GPA: 18.17/20, Distinction: *With High Honors*

PUBLICATIONS AND PREPRINTS

NeurIPS Effective Context in Transformers: An Analysis of Fragmentation and Tokenization — 2026
*A. Fesharaki, **M. Rami**, A. Tchamkerten*

Preprint Zero-Error Functional Source Coding with Limited-Lookahead Side Information — 2026
*A. Fesharaki, **M. Rami**, A. Tchamkerten*

RESEARCH EXPERIENCE

Institute of Science & Technology Austria (ISTA) — Klosterneuburg, Austria
Scientific Researcher — Supervisor: Prof. Marco Mondelli — Nov 2026 – Present

- Theoretical analysis of the privacy–generalization trade-off and membership inference attacks in high-dimensional, overparameterized learning models.

Télécom Paris — Paris, France
Research Intern — Supervisor: Prof. Aslan Tchamkerten — March 2026 – Oct 2026

- Effective Context in Transformers: An Analysis of Fragmentation and Tokenization — theoretical analysis of how data representation fundamentally affects the achievable loss of finite-context next-token prediction models (e.g. Transformers).
- Zero-Error Functional Source Coding with Limited-Lookahead Side Information — characterizing fundamental limits of zero-error functional source coding with causal and limited-lookahead side information at the decoder.

MCI (Mobile Communications of Iran) — Tehran, Iran
AI Research Intern — Supervisor: Dr. Reza Kazemi — June 2023 – Sept 2023

- Post-training and fine-tuning of language encoders (BERT-based architectures) for Persian language for certain down-stream classification tasks.

HONORS & AWARDS

- Ranked **1st** in MSc program cohort, Institut Polytechnique de Paris
- Institut Polytechnique de Paris MSc Scholarship
- Ranked in the **top 15%** among Electrical Engineering undergraduate cohort, Sharif University of Technology
- Ranked in the **top 0.05%** of participants in the Iranian National University Entrance Exam

TEACHING EXPERIENCE

Teaching Assistant — Sharif University of Technology

- Linear Algebra
- Engineering Mathematics
- Deep Learning
- Information Theory

WORK EXPERIENCE

Data Scientist @ PropertyGPT: LLM tool call development — Oct 2024 - Sept 2025
Software Engineer @ MCINext: Scalable API development — Jan 2024 – Sept 2024